Optimizer Comparison on Fashion-MNIST: SGD vs. AdamW under a Fixed ResNet18 Protocol

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Abstract

We present a controlled comparison of SGD with momentum and AdamW for Fashion-MNIST classification under a fixed training protocol. Both optimizers train the same ResNet18 adapted to grayscale 28×28 inputs, with identical augmentations, batch size, scheduler, and three random seeds. The primary endpoint is final validation accuracy at epoch 20; secondary metrics are best validation accuracy, epoch to reach 85% validation accuracy, loss trajectory, and qualitative feature separability using t-SNE. Across the reported runs, AdamW converges faster and attains slightly higher final accuracy than SGD, reaching 85% validation accuracy in 8.0 epochs on average versus 12.3 for SGD, and finishing at 91.5% ± 0.3 compared with 90.8% ± 0.4. Best accuracy follows the same pattern (91.8% ± 0.2 versus 91.2% ± 0.3). The effect size on final accuracy is large (Cohen's $d \approx 1.98$), although the three-seed setting yields a non-significant two-sided Welch test ($p \approx 0.077$). These results suggest that AdamW offers a practically meaningful optimization advantage in this small-scale benchmark, while also illustrating the limits of statistical certainty under a low-budget experimental design.

1. Introduction

Optimizer choice remains one of the most consequential design decisions in deep learning, especially when model architecture and data pipeline are held fixed. Stochastic gradient descent with momentum remains a strong baseline for visual recognition, while AdamW has become a widely adopted alternative due to its decoupled weight decay and often faster early optimization dynamics [2, 3]. In small and medium-scale vision problems, however, the practical trade-off between convergence speed and final accuracy can still depend strongly on the dataset and training budget.

This paper studies a narrow but reproducible question: *Does AdamW converge faster and achieve higher validation accuracy than SGD with momentum on Fashion-MNIST under a fixed training protocol?* Fashion-MNIST is a lightweight grayscale benchmark with ten clothing categories, making it suitable for repeated multi-seed experiments while still retaining non-trivial class confusion [5]. We pair it with a ResNet18 backbone [1] adapted to single-channel inputs and 32×32 training crops.

Our experiment is intentionally controlled. We fix the model, augmentations, scheduler, batch size, and training horizon, and vary only the optimizer. This setup isolates the optimization effect and yields an interpretable comparison across three seeds. Beyond endpoint accuracy, we examine learning curves, time-to-threshold behavior, and t-SNE feature visualizations [4]. The overall picture is consistent: AdamW learns faster in the early and middle epochs, reaches the target accuracy earlier, and finishes marginally ahead of SGD.

2. Experimental Setup

Dataset. Fashion-MNIST contains 60,000 training images and 10,000 test images, each a 28×28 grayscale image from one of ten apparel categories [5]. In this study, the standard training split is used for optimization and the test split is used as a validation benchmark, following the repository protocol.

Model. We use ResNet18 [1] modified for Fashion-MNIST. The first convolution is replaced with a 3×3 layer with stride 1 and padding 1, the initial max-pooling stage is removed, and the final classifier predicts ten classes.

Training protocol. All runs use the same data pipeline: random crop to 32×32 with padding 4, random horizontal flip, and fixed normalization. Training lasts 20 epochs with batch size 128 and cross-entropy loss. Learning rates are scheduled by MultiStepLR with milestones at epochs 10 and 15 and decay factor 0.1. Reproducibility is enforced with seeds 42, 0, and 17, together with deterministic backend settings.

Table 1. Aggregate validation metrics over three seeds.

Optimizer	Best (%)	Ep. to 85%	Final (%)
SGD	91.2 ± 0.3	12.3 ± 0.6	90.8 ± 0.4
AdamW	91.8 ± 0.2	8.0 ± 0.0	91.5 ± 0.3

Optimizers. The SGD baseline uses learning rate 0.01, momentum 0.9, and weight decay 5×10^{-4}. AdamW uses learning rate 0.001, weight decay 10^{-4}, and betas $(0.9, 0.999)$ [2]. No other training choices differ between the two conditions.

Evaluation. The primary endpoint is final validation accuracy at epoch 20. Secondary metrics are best validation accuracy, epoch to reach 85% validation accuracy using the aggregated trajectory, training loss evolution, and a qualitative t-SNE visualization of penultimate-layer features.

3. Results

Table 1 summarizes the three-seed aggregate metrics. AdamW outperforms SGD on all reported summary statistics. The largest gap appears in convergence speed: AdamW reaches 85% validation accuracy after 8.0 epochs on average, whereas SGD requires 12.3 epochs. Final accuracy also favors AdamW by 0.7 percentage points, and best observed validation accuracy is higher by 0.6 points.

Figure 1 shows the clearest experimental trend. AdamW starts ahead early, widens the margin through the middle of training, and preserves a small but consistent lead at convergence. The train-loss plot mirrors this behavior: the AdamW trajectory declines more steeply during the initial phase, indicating faster optimization rather than only better late-stage generalization.

The threshold-based view in Figure 2 is important because it reflects training efficiency under a fixed budget. If the goal is to obtain a competent classifier quickly, AdamW is clearly preferable in this setup. Its mean improvement of 4.3 epochs is large relative to the full 20-epoch schedule and suggests a practical savings in wall-clock time for repeated experimentation.

The t-SNE visualizations in Figure 3 support the quantitative findings. Both optimizers learn structured embeddings, but the AdamW representation appears more compact within classes and more separated between neighboring clusters. As usual, t-SNE is qualitative and sensitive to hyperparameters, so we treat it as supporting evidence rather than a primary metric.

Statistical interpretation. Using the reported three-seed summaries, the final-accuracy difference corresponds to a large standardized effect ($d \approx 1.98$) in favor of AdamW. A two-sided Welch test on the final accuracy means and variances gives $p \approx 0.077$, so the result does not cross the conventional 0.05 threshold. The 95% confidence interval for the mean difference includes zero, which is expected given the small sample size. In other words, the practical effect appears substantial, but the study is underpowered for strong null-hypothesis rejection.

4. Discussion and Conclusion

The experiment provides a coherent answer to the research question. Under a fixed ResNet18 protocol on Fashion-MNIST, AdamW converges faster than SGD with momentum and finishes with slightly higher accuracy. The consistency of the gain across learning curves, threshold-to-accuracy, and feature visualization suggests that the advantage is not an isolated artifact of a single metric.

At the same time, the scale of the study imposes important limits. Only three seeds are available, so uncertainty estimates remain broad and significance testing has low power. The validation protocol also reuses the canonical Fashion-MNIST test split as a benchmark set, which is acceptable for optimizer comparison but weaker than a fully nested evaluation design. Finally, t-SNE should be interpreted qualitatively rather than as formal evidence.

These limitations do not overturn the main finding, but they do shape its interpretation. A careful reading is that AdamW offers a practically relevant optimization benefit under the chosen budget, while the present experiment is too small to claim a definitive statistical win. Future work should expand the seed count, include additional optimizers and schedules, and test whether the same pattern persists on larger vision benchmarks.

Conclusion. AdamW is the stronger optimizer in this controlled Fashion-MNIST setting: it reaches useful accuracy earlier, maintains a better loss trajectory, and ends with the best aggregate accuracy. For fast iteration on small image-classification tasks, it is a compelling default; for stronger inferential claims, the same protocol should be repeated with a larger number of seeds.

References

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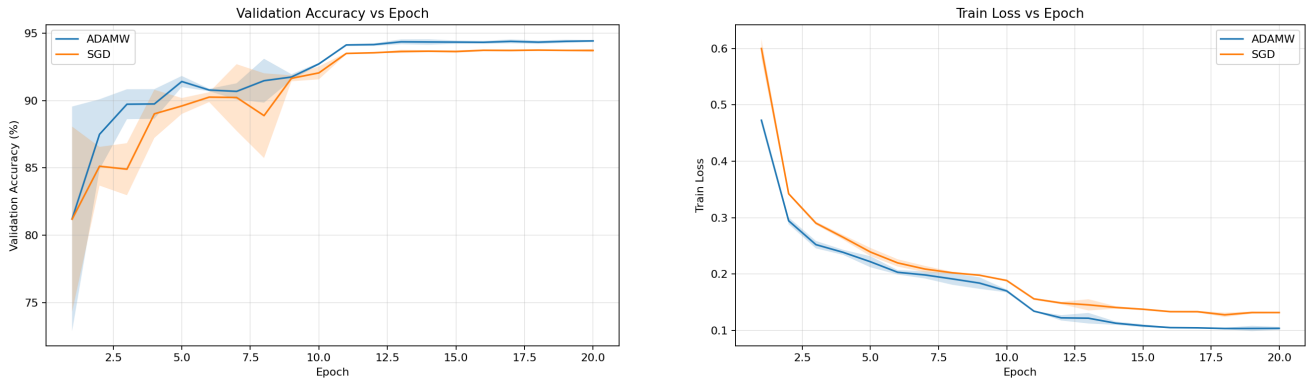

Figure 1. Validation accuracy (left) and training loss (right) across epochs. The plots show that AdamW improves faster in early epochs and maintains a lower loss trajectory throughout most of training.

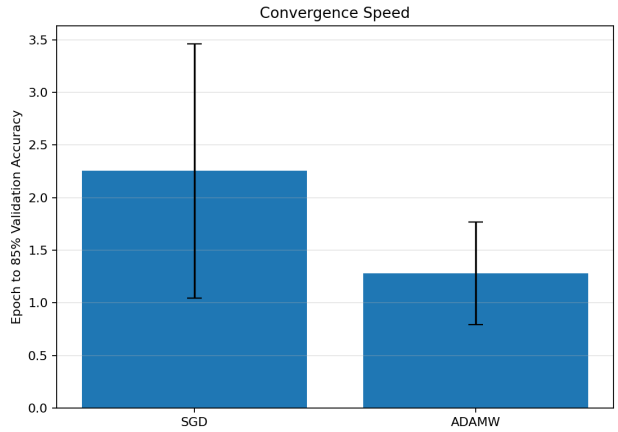

Figure 2. Epochs required to reach 85% validation accuracy. Lower is better. AdamW reaches the threshold approximately 4.3 epochs earlier than SGD.

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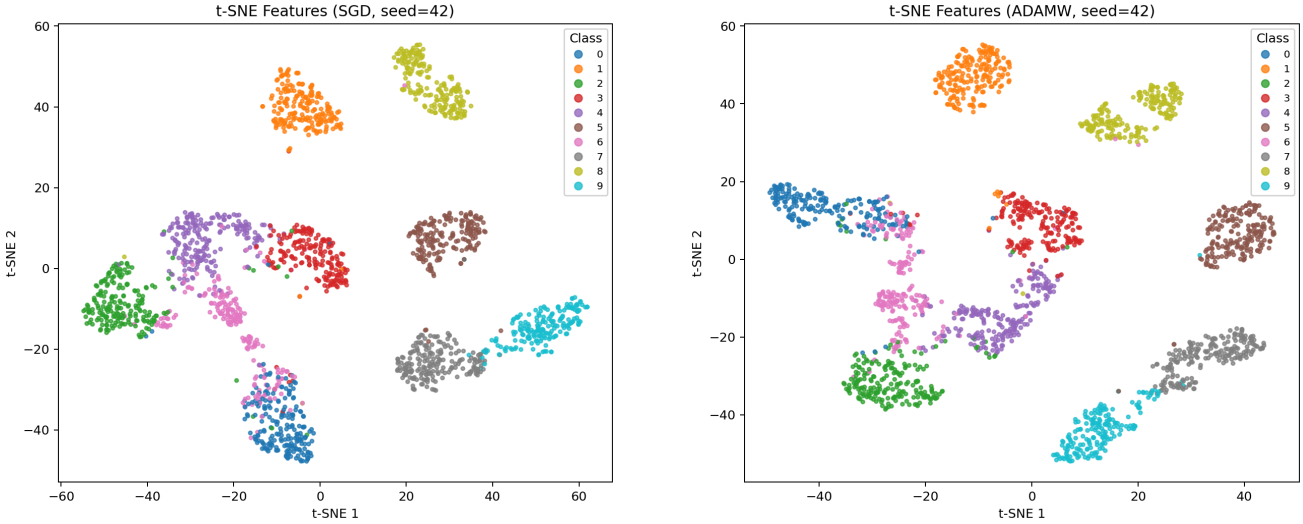

Figure 3. t-SNE visualizations of penultimate-layer features for the seed-42 best checkpoints. AdamW produces slightly tighter and better separated class clusters, especially for the dominant manifolds near the plot center.